

## Particle Swarm Module Homework

**Due March 27 Midnight**

For the eggholder function (next page) do the following:

1. Code a simple full model PSO with the star neighborhood (global neighborhood – all solutions are neighbors of all others) to solve the problem. To do this you need to encode the problem as two real variables with associated  $\mathbf{p}$  and  $\mathbf{v}$  vectors, and select values for  $\varphi_1$  and  $\varphi_2$ . Do not include inertia nor constriction in this PSO. Devise a way to deal with the boundary constraints.
2. Run your PSO for five different seeds.
3. Perform the following changes on your PSO code, one at a time, and compare the results.
  - Try a cognition only PSO ( $\varphi_2=0$ ) and run for the same 5 seeds.
  - Try a social only PSO ( $\varphi_1=0$ ) and run for the same 5 seeds.
  - Try a selfless model ( $\varphi_1=0$  and  $g \neq i$ ) and run for the same 5 seeds.
  - Try the ring topology (each solution has two neighbors, however you wish to calculate these) using the full model (from number 1 above) and run for the same 5 seeds.
  - Add the inertia factor to the original PSO (full with star neighborhood) and run for the same 5 seeds.
  - Add the constriction coefficient to the original PSO (full with star neighborhood, no inertia) and run for the same 5 seeds.
  - Now include both the inertia and the constriction factors in this PSO and run for the same 5 seeds.
  - Vary the number of particles with the PSO from just above (full with star neighborhood with constriction and inertia) to higher and lower (one value of each) and run each for the same 5 seeds.

This should result in  $5+5+5+5+5+5+5+5+10 = 50$  runs.

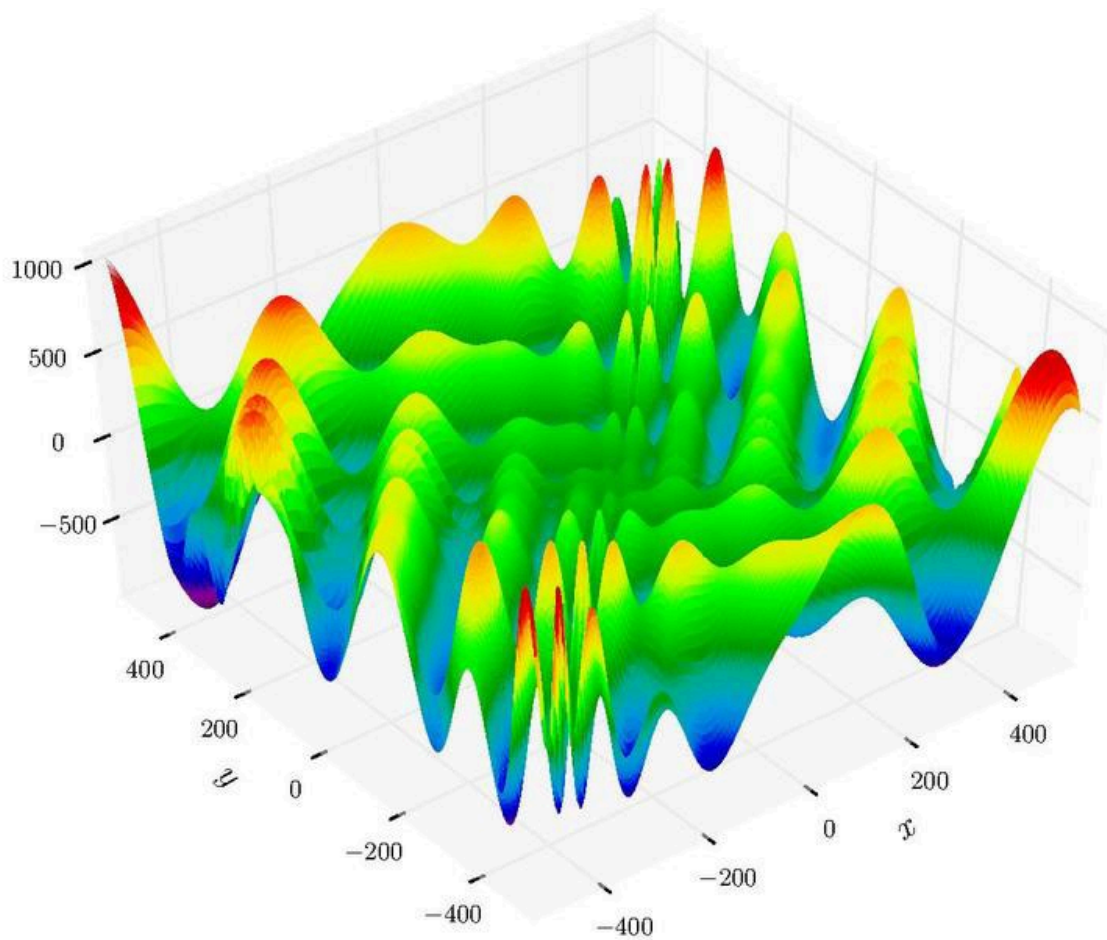
4. Please include your code with your homework and summarize your results.

## Eggholder function - Minimization

$-512 < x_1, x_2 < +512$

$$f(\mathbf{x}) = -(x_2 + 47) \sin \left( \sqrt{\left| x_2 + \frac{x_1}{2} + 47 \right|} \right) - x_1 \sin \left( \sqrt{|x_1 - (x_2 + 47)|} \right)$$

$$f(\mathbf{x}^*) = -959.6407, \text{ at } \mathbf{x}^* = (512, 404.2319)$$



Extra Problem – up to 15 points extra on this homework

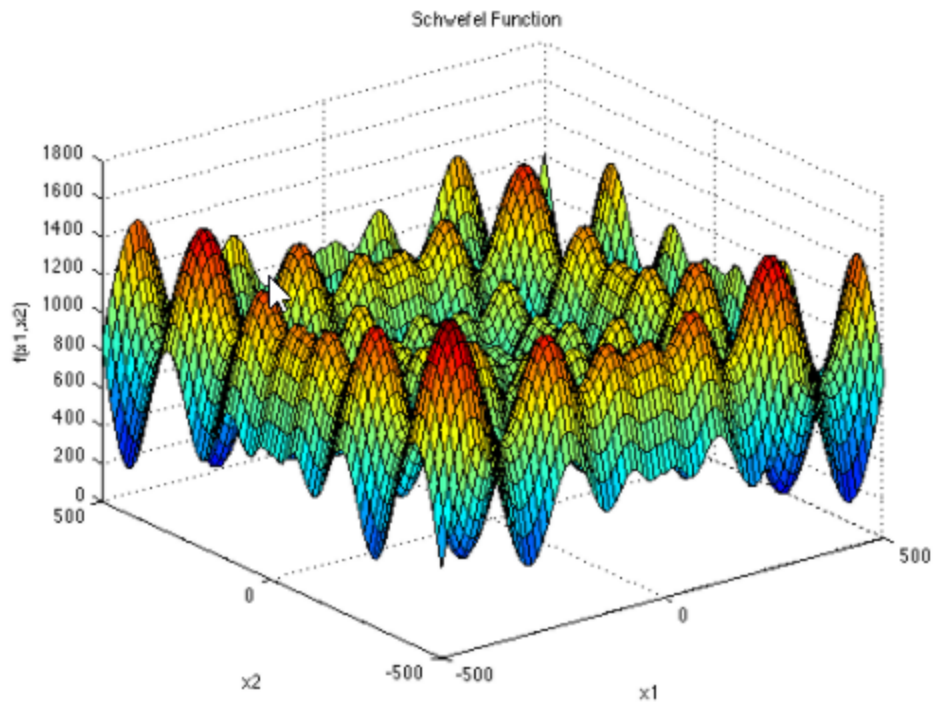
Do the same that you did for the eggholder problem for the minimization problem below for four dimensions ( $n = 4$ ). This is the Schwefel function.

$$f(x_1 \cdots x_n) = \sum_{i=1}^n (-x_i \sin(\sqrt{|x_i|})) + \alpha \cdot n$$

$$\alpha = 418.982887$$

$$-512 \leq x_i \leq 512$$

minimum at  $f(420.968746, 420.968746, \dots, 420.968746) = 0$



The two-dimensional version of this function.